Rigour check: the two-level biofilm model from one variational principle

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This note lists, for the model of `unified_model_derivation.tex`, what is derived, what is checked by machine, what is checked by computation, what is assumed, and what is still open. Every entry names the file that produces it. All files are in `JAXFEM/` of the repository unless stated otherwise.

1 Derivation

Statement	Status	How it is established
The discrete equations solved are the Euler–Lagrange equations of the discrete functional $E_h + \Delta t\, D_h + C_h$	established	The residual is the automatic derivative (JAX) of the functional; no equation is written by hand (`unified_two_level_1d.py`, `functional`, `residual`).
The point model of Klempt et al. 2026 is the stationarity condition of the energy Ψ_{pm} and dissipation D_{pm} used here	established	One node, no coupling: the composition rows equal the repository's point-model residual (`core_hamilton_1d.residual`) to round-off, every row (`unified_two_level_variational_check.py`).
The hand-written residual of the first version agrees with the automatic one	established, with one known difference	Composition rows to round-off; the phase row differs on purpose (the hand version approximated one term and let the barrier terms act on the phase).
Scheme (c) of the element is the limit $\lambda \to 0$, $\beta_s = 0$ of the model	established by inspection	The composition equations lose the coupling terms and the expansion equation becomes $\dot{\alpha} = k_\alpha \phi \sum_i s_i \psi_i$ (derivation note, Section 4).
$\frac{\mathrm{d}}{\mathrm{d}t}\int E = -2\int D \le 0$ without external power	proved for the model with the barrier terms as stored energy	Derivation note, Section 5. For the version with the barrier terms as a constraint the inequality holds in the limit $K \to 0$; a discrete check is open.
The front motion follows from the growth kinematics	established in 1D, 2D and 3D by computation	Below.

2 Checks by computation

Check	Result	File
Uniform field, no coupling: every node equals the point model solved to convergence	8.9×10^{-11} (two-level), 1.6×10^{-13} (one-level)	`unified_two_level_1d.py`, `unified_1d_monolithic.py`
Energy decrease of the one-level model, every step	largest change -5.6×10^{-3}, no increase except at the antibiotic switch (external input)	`unified_1d_monolithic.py`
Energy decrease of the fuel model with closed boundary	largest change -9.8×10^{-3}, no increase	`unified_two_level_fuel_1d.py`
Newton convergence	residual $< 10^{-8}$ every step, at most 31 iterations, adaptive halving of the step when needed	`unified_two_level_1d.py`
Time step, two-level 1D	60 against 120 steps: $\alpha - 1$ changes by 0.7 %	`unified_two_level_1d.py`
Mesh, 3D lumped	16^3, 24^3, 32^3: edge displacement within 3 %	`fuel_front_3d.py`
Mesh, 3D five species	16^3 against 24^3: within 8 % (constant c^*), 9 % (local nutrient)	`fuel_front_3d_species.py`
Point model inside the 3D runs solved to convergence	residual $\le 4.5 \times 10^{-13}$ every step	`fuel_front_3d_species.py`

3 Assumptions, each with its reason

Assumption	Reason	Tested
Two levels (phase ϕ and composition s_i, s_0, ψ_i)	One level fills the outer liquid from the barrier floor (`unified_1d_monolithic.py`); the two voids are different objects	yes
Weight $h(\phi) = \phi^2$ of the composition energy	With $h = \phi$ the composition force does not vanish in the outer liquid and it fills again; ϕ^2 is the weight of the mechanical term of Klempt et al. 2024	yes
The barrier terms act as a constraint, not as stored energy	Their value (517 at the calibration's start against 8 at $\psi = 0.5$) would make dying cells grow the biofilm	yes
$\phi \leq 1$ by a one-sided barrier $K_\phi/(1-\phi)^2$	The phase equation has no upper bound (Klempt et al. 2024 neither); ϕ reached 2.7 at large k_α	yes
Growth as a nutrient reaction (Onsager), the living biomass as catalyst	A growth energy on the living biomass keeps ψ up at large growth and defeats the antibiotic; the catalyst form has no force on ψ and makes the total energy decrease	yes

Assumption	Reason	Tested
Ideal-solute free energy $c(\ln c - 1)$ of the nutrient, linear kinetics $\dot{\xi} = L(Y \ln c - \mu_b)$	Simplest forms consistent with the principle; $\mu_b = Y \ln c_{\text{stop}}$ gives a stop concentration	sensitivity only
λ (energy scale of the composition), β_s, L_0, Y, c_{stop}, D	Not from a paper; λ is the one that matters for the antibiotic's effect on the phase, β_s does not matter for a seed	λ, β_s: yes; others: no
Time link of the composition ($s = 0.2375$ per T^*)	The calibration's link; in the model it is the ratio of dissipation coefficients and can be fixed by the measured volume	no
Local nutrient in the point model: interaction $c^* c(\mathbf{x})/c_{\text{ref}}$, $c_{\text{ref}} = 1$	The element's local-nutrient option; the relation of c^* to the field's c should come from the nutrient level of the calibration	no
Front term of Klempt et al. 2024 left out	Not derivable from a potential (their Remark 1); the edge moves towards the nutrient by the growth kinematics alone	yes (1D, 2D, 3D)
Mechanics in the 2D/3D checks: small strain, $\nu = 0.3$, linear elasticity, nutrient in the reference configuration	Checks of the kinematics only; the full model uses Klempt et al. 2024's mechanics ($\nu = 0.49$)	no
Lumped composition in `fuel_front_2d.py`, `fuel_front_3d.py`	Kinematics check; replaced by the five-species point model in `fuel_front_3d_species.py`	yes

4 Open

1. A discrete energy check of the two-level model with the barrier terms as a constraint.
2. The limit $\phi \to 0$ of the composition equations (regularised by ε).
3. The composition acting on the phase ($\lambda > 0$) and the composition diffusion (β_s) in 3D; the 3D runs so far use the decoupled limit.
4. Finite strain and $\nu = 0.49$ in the 2D/3D checks; nutrient diffusion in the current configuration.
5. Existence and uniqueness of solutions (not attempted).
6. A reader other than me: the derivation has not been checked by Dr. Soleimani or anyone else.
7. Overlap with F. Klempt's unpublished two-species work: only published sources were used; to be discussed before any publication.

5 What the checks do not cover

The checks establish that the equations are the ones the energy and dissipation imply, that the solver solves them, and that the results do not depend on the mesh or the time step at the level stated. They do not establish that the model is right: the choices in Section 3 are modelling assumptions, and the only data used so far are the calibrated interactions of the point model. The measured volumes (Heine et al.) and an antibiotic experiment would test the growth law and λ.